Solution Requirements

Date: 06 October 2026 **Team ID:** unm293200909 **Project Name:** FitBuddy-AI fitness plan generator using Gemini models

Maximum Marks: 4 Marks

Assigned Team Member: Kavyasri.A

Define Functional and Non-Functional Requirements: Create a comprehensive list of requirements to define exactly what the FitBuddy solution will do and how well it will perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority	Assigned Work
1	Authentication	Users securely sign in or verify identity before accessing personalized fitness-plan features.	High	Kavyasri.A – Define login/verification flow and user access requirements.
2	Authorization levels	Define user roles and permissions, such as user access to personal plans and authorized administrative access.	Medium	Kavyasri.A – Specify roles and permissions.
3	External interfaces	Integrate with Gemini models/API to generate personalized fitness plans based on user-provided information.	High	Kavyasri.A – Document Gemini API and required external interfaces.
4	Transactions processing	Process user inputs, send suitable prompts to the Gemini model, receive generated recommendations, and store/display the result.	High	Kavyasri.A – Describe input-to-plan processing workflow.
5	Reporting	Provide generated fitness plans and useful summaries of workouts, goals, schedules, and recommendations.	Medium	Kavyasri.A – Define plan summaries and reporting requirements.
6	Business rules, etc.	Generate plans according to user goals, fitness level, preferences, available schedule, and provided constraints.	High	Kavyasri.A – Define personalization rules and constraints.
7	Compliance to laws or regulations	Protect personal information and apply appropriate privacy and responsible-use practices for fitness-related user data.	High	Kavyasri.A – Document privacy and responsible-data requirements.
8	Other	Allow users to view, understand, and use the AI-generated fitness plan in a simple and organized manner.	Medium	Kavyasri.A – Define additional user-facing requirements.

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria	Assigned Work
1	Performance & Speed	The system should respond quickly when users submit their fitness details and request a plan.	Typical page interaction and response target: under 2–3 seconds excluding external AI/API delays.	Kavyasri.A – Define performance expectations.
2	Scalability	The application should support growth in users, stored plans, and AI requests without major performance degradation.	Architecture should support increasing users and data with scalable backend/API resources.	Kavyasri.A – Define scalability needs.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria	Assigned Work
3	Security & Data Privacy	Protect account information and fitness-related personal data using secure authentication, controlled access, and encrypted communication.	HTTPS/TLS for data in transit; secure credential handling; least-privilege access.	Kavyasri.A – Define security and privacy criteria.
4	Reliability & Availability	The system should remain available and handle temporary AI/API failures gracefully.	Target high availability; show a useful error/retry message when the Gemini service is unavailable.	Kavyasri.A – Define reliability and failure-handling criteria.
5	Usability & Accessibility	The interface should be simple, readable, mobile-friendly, and easy for users to enter details and understand generated plans.	Clear navigation, readable content, responsive layout, and accessible form controls.	Kavyasri.A – Define usability/accessibility criteria.
6	Other	Maintainability and portability should allow the application to be updated as AI models and requirements evolve.	Modular code, documented APIs/configuration, and easy replacement/update of model integration.	Kavyasri.A – Define maintainability and portability requirements.

Work Assignment Summary: Kavyasri.A is assigned the requirements work in the same order as the provided template: FR 1–8 followed by NFR 1–6.